

On Optimising μ SR Sample Design and Mounting using Monte Carlo Simulations

Technical Report

Science & Technology Facilities Council

Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX



Science and
Technology
Facilities Council

ISIS Neutron and
Muon Source

Author: Debjyoti Sengupta, Peter J Baker

A digital twin-based Design Search and Optimisation (DSO) Process for sample mounting in μ SR experiments using Monte Carlo simulations

ISIS Neutron and Muon Source,
STFC Rutherford Appleton Laboratory

Abstract:

Optimising the sample mounting to stop the muon beam is helpful when planning experiments with limited sample quantity. Using the instrument range curve is a good start in simple cases but computational tools can be used to help for more complex cases. We have established a design, search, and optimisation (DSO) process using musrSim (based on Geant4 & ROOT) and SRIM/TRIM software tools to help in planning complex experiments. This has been tested against experimental data to confirm the comparison with the simulations, based on the Monte Carlo algorithm. Comparisons have also been made with the methods previous proposed by Ziegler and Elson et al. To explain the method to new muon users we are also preparing an online training course that will be available soon.

Keywords:

DSO, Monte Carlo, Muon Instruments, Muon Beam, Sample Design, μ SR, SRIM/TRIM, musrSim

Commented [BP(1)]: Needs to be snappier and focus on what the reader will learn by reading the report, and how they can make use of the results contained.

Commented [BP(2)]: As for the title. What in specific terms, Why in general terms, How briefly.

Table of Contents

Introduction	4
Muons	4
Muon Spectroscopy	4
μ SR	4
Hardware	5
Muon Beamline Design	5
Sample Environment.....	6
Software.....	8
Geant4 & ROOT.....	8
musrSim & musrSimAna	9
SRIM & TRIM	11
Muon Experiment Decision & Optimisation Process (MEDOPS)	15
Literature Review & Solution-Neutral Problem Statement	15
Decision Flowchart.....	17
Process Methodology	18
Process Verification & Validation – Sample Design & Muon Implantation Study	20
Experimental Outcomes, Planning and Setup	20
Results and Data Analysis	23
Conclusion.....	34
Author Contributions	34
Acknowledgements.....	34
Appendix	34
Appendix A: Essential Instrument and Sample Environment Information	34
Appendix B: Setup Variables	34
Appendix C: musrSim Guide	35
Appendix D: SRIM/TRIM Guide	37
References	39

Introduction

Muons

Muons (denoted as $\mu^\pm$) are fundamental particles that are characterized by properties reminiscent of a heavy electron. They possess a unit charge which is positive for positive muons (μ^+) and negative for negative muons (μ^-), have a mass 200 times the mass of an electron (known as the Goldilocks Mass), and they are highly unstable in nature with a lifetime of approximately 2.2 μ s. According to the Standard Model of Particle Physics, a muon is classified as a charged lepton, alongside electron e^- and the tau particle τ^- . They all have a spin value of -1/2.

Muons very readily decay into lighter electrons due to their unstable nature. The muon-to-electron conversion process can be illustrated in the following equations:

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$

$$\mu^+ \rightarrow e^+ + \bar{\nu}_\mu + \nu_e$$

Muons decay briefly into bosons, which are highly unstable, bosons then quickly decay into electrons. Lepton number conservation dictates that this process must produce a charged electron species and a neutrino/antineutrino ($\nu/\bar{\nu}$) pair. Neutrinos are also leptons without any electrical charge. The production of the charged species also fulfils the requirement for the charges to balance; the nature and charge of this species is dependent on the starting muon. While neutrinos are difficult to detect, charged particles like muons and positrons and electrons can be easily detected by using appropriate techniques.

Muon Spectroscopy

The ISIS Muon Group operates six muon spectrometers that use both positive and negative muons. These spectrometers are used by researchers for a wide range of studies like elemental analysis, modelling hydrogen behaviour, magnetism & superconductivity and much more.

μ SR

μ SR (Muon Spin Rotation/Relaxation/Resonance) is an experimental technique that measures any changes in muon spin polarisation over time by recording the direction of muon decay products after implantation. The changes in muon spin reflect the physical processes happening in the sample. This can be used to measure magnetic fields inside samples for studies related to magnetism, semiconductors and superconductivity, and examine chemical processes. The direction in which the resultant positrons/electrons are emitted reflects the direction of the spin of the original muon. Changes in muon spin polarisation can, therefore, be measured as changes in the rate of positron emission using stationary detectors at different angles around the sample.

The experimental technique can be further categorised based on their principles, each having different applications. The most frequently used techniques are TF- μ SR, ZF- μ SR and LF- μ SR, which vary on how the external magnetic field is applied to the muon spin. TF- μ SR (Transverse Field μ SR) requires the field to be applied transverse (perpendicular) to the muon spin while LF- μ SR (Longitudinal Field μ SR) requires the magnetic field applied to the system parallel to the direction of the spin of the muons. ZF- μ SR can be performed by not applying any magnetic field to the system, hence the name (since there is Zero magnetic Field). Besides the above techniques which use positive muons, negative muons can be used for μ SR experiments focussing on aspects of elemental analysis. Other techniques make use of sample environment like pressure cells, or other external stimuli like RF, or changing the muon beam energy (for slow μ SR), or changing data analysis like FT- μ SR (Fourier Transform) where the data is numerically translated into a frequency.

Hardware

Muon Beamline Design

Muon beams are produced from a high energy proton beam which interacts with a sheet of carbon. This then creates charged pions which rapidly decay into muons. The muon beams is then manipulated using magnets, directing towards the instruments shown in figure x.

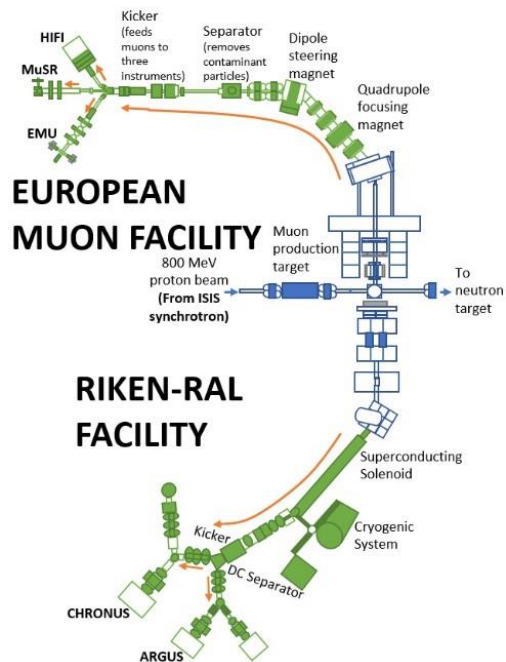


Figure 1: Beamlines at ISIS-RAL

Surface muon beams are emitted from pions decaying in the surface layer of the pion production target. They can have 100% spin polarisation, with partial polarisation possible for other types of beam. This means that samples do not need to be isotopically enriched or have large magnetic fields applied to them, unlike in NMR (Nuclear Magnetic Resonance) and ESR (Electron Spin Resonance).

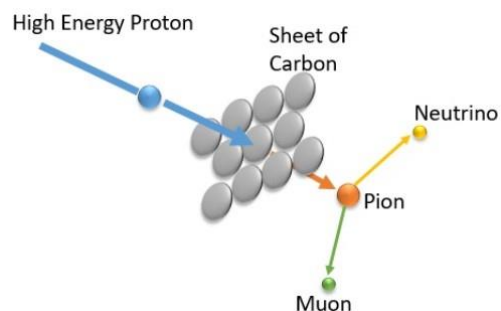


Figure 2: Illustration of muon production from high energy proton beam colliding with the production target

Generally, most muon beams are surface muon beams. However, beams can also be produced from pions decaying in flight - decay muon beams. In these beams, muons are emitted both forward and backward of the pion rest frame. The “backwards” muons have significantly different momenta from the “forwards” ones and can be selectively extracted by a bending magnet. The disadvantages of backwards muon beams are their higher momentum, larger momentum spread and larger phase space. However, these “backwards muons” are the only way to investigate samples that are contained in thick target vessels such as gases, liquids and sample in high-pressure effect studies.

For some experiments lower energy muons are required. For example, lower energy muons are sometimes needed to investigate thin layer or multilayer samples, because they can be stopped nearer the sample’s surface. These lower energy muons are often referred to as “sub-surface” muons. They still have almost 100% spin-polarisation and are produced to almost the same numbers as surface muons. Muon beams have been developed with energies of 0.5-30keV and implantation depths of less than 1nm in a solid sample. Much research has been done into the development of “ultra-slow” muon beams. However, currently the most successful attempt at this has been through simple moderation in a thin degrader made of nitrogen or argon gas. The “moderated” muons are still highly polarised; but their intensities are reduced to several orders lower than that of surface muons.

Typically, muon beam energies allow the muons to penetrate up to around 0.1mm into materials, depending on the density. For deeper implantation decay muons can be used. However, to stop muons in a thinner sample “moderated” muons can be used- ultimately allowing controlled implantation down to 5-300nm from the surface. Surface muon beamlines typically have an estimated low energy of ~ 4 MeV and corresponding momentum of ~ 28 MeV/c, while decay muon beamlines have a typical momenta of 40 to 120 MeV/c.

Sample Environment

Muons enter and stop inside samples; they do not scatter from, diffract or reflect off. Once entering the sample, the muons implant themselves amongst its original atoms, as shown in Figure x. When inside a sample, muons can act as probing particles and gain information. The information coming from the muons is in the form of their spin polarisation and is measured as a function of time after implantation. This is measured using their decay into positrons, which are emitted asymmetrically relative to the spin polarisation. When inside a material, muon spin polarisation changes according to the processes going on inside, changing the measured signal.

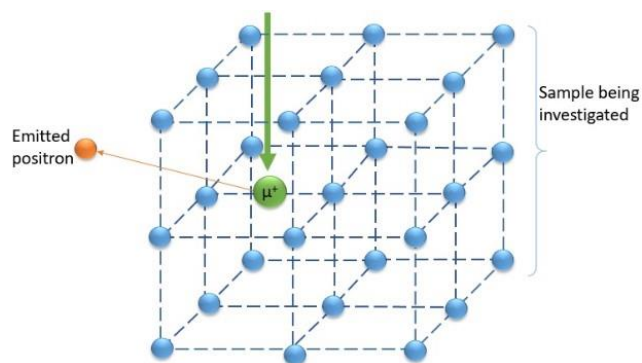


Figure 3: Muon implantation in sample

Commented [BP(3)]: Please add a short section about how muons stop in materials here.

Commented [BP(4R3)]: Include what range curves are and how they come about.

Commented [BP(5)]: Needs something on how samples are mounted in different ways because that is what is being simulated.

Commented [BP(6R5)]: Also needs something on the range of sample environment that is used and how this impacts on how muons stop in the sample.

In most cases muons implant into samples without loss of spin-polarisation. Implantation of muons can cause ionisation of atoms and scattering of electrons within a sample. Muons rarely replace an atom in the sample. They usually stop at interstitial sites of crystals or addition sites of molecules.

Sample mounting is one of the critical factors a user needs to consider during the planning stages of an experiment. The underlying principles behind mounting remain the same despite differences across facilities and instruments. Ideally, a sample holder made of element with small nuclear moments is usually used to minimise the muon polarisation relaxation. However, if the holder is made of element with large nuclear moments (due to ease of machining and cost reduction), the data can be corrected by adding the number of muons stopping in the holder rather than the sample. Typical holders maybe creating a silver foil packet, squares of aluminium or titanium with shapes milled out for powdered sample to sit.

Examples for other methods for mounting could be the use of small cells and cryostats for air-sensitive samples and liquids, the use of large sample cells for gaseous samples, and the use of different pressure cells where large amounts of sample is contained in the cell within a pressure medium through a gas or oil. The resultant relaxation rate of muons for these cells under such circumstances could be dependent on temperature.

The mean stopping power for high-energy muons in a sample made of a given material can be stated in the following expression:

$$\langle -dE/dx \rangle = a(E) + b(E)E$$

Where E is the total energy, $a(E)$ is the electronic stopping power and $b(E)$ is radiative stopping power due to processes like photonuclear interactions, pair production and bremsstrahlung.

$$b \equiv b_{\text{bremsstrahlung}} + b_{\text{pair}} + b_{\text{nuclear}}$$

Usually, $b(E)E$ is less than 1% of $a(E)$ for beam energies less than or equal to 100 GeV for majority of materials.

It is important that the muons stop within the sample rather than the cryostat or the mounting apparatus. This can be done by obtaining the correct predictions for degrader thickness or areal mass density of a given sample by examining the range curve, illustrated in Figure x. Range curves are an essential piece of information a user must refer to from the user manual for a given μ SR instrument. The interpretation of the range curve is rather simple – ensure that the sample centre is at the midpoint of the curve. We must note that the curves can vary for different environments used for mounting.

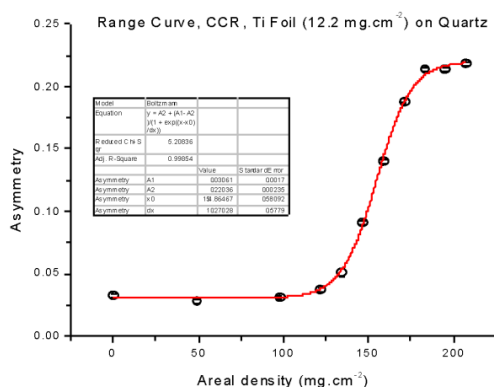


Figure 4: Range curve for a cryostat (sample environment)

Software

The Muon Experiment Simulation Software Cycle (MESSC) has been illustrated shown using Figure x. The software involved in performing simulations of ISIS muon experiments are MuSRSim & MuSRSimAna, which are based on the CERN-developed Monte Carlo-based toolkit Geant4 and data analysis toolkit ROOT. SRIM & TRIM are also part of this package as they help the user with sample design calculations and experiment simulations. We have identified other software like IBEX, ANSYS and more, which could be integrated into the larger cycle such that the simulation provides more data that the user may find useful before they arrive for their experimental run at ISIS Neutron and Muon Source.

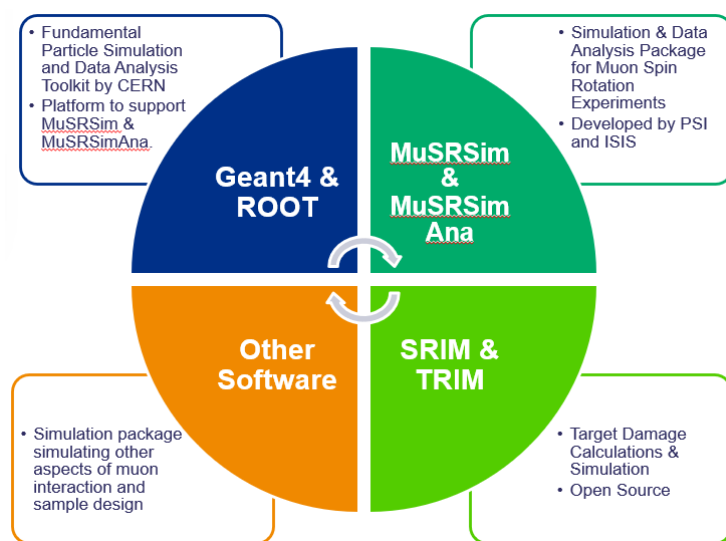


Figure 5: Muon Experiment Simulation Software Cycle

Geant4 & ROOT

Geant4 (often referred to as G4) is a C++ based open-source toolkit developed by CERN, Geneva to perform Monte Carlo simulations of particle interactions (strong and weak), movement and radiation through matter. Developers have used Geant4 for developing applications that are able to simulate all areas of particle transport. The toolkit serves as the first half of important software packages essential to run *muSRSim* & *muSRSimAna*. Many of the functionalities of *muSRSim* are the direct derivatives of the G4 toolkit. Some of them can be listed as follows:

- Create a model of the μ SR instrument geometry with shapes and material definition, and GUI visualisation,
- Particle navigation tracking,
- Physics interactions and generation of secondary particles.

The G4 toolkit has found multiple applications across various domains. Areas can cover fields of research like High Energy and Nuclear Physics, medical physics and space science. Various HEP and nuclear physics experiments have used the toolkit successfully for design and optimisation of instruments and detectors, whilst creating a digital twin model of the experiment which can be used to generate simulated events as part of the data analysis process.

Commented [BP(7)]: For each software tool, a paragraph on what it does and any physics results needed to understand that should be in the introduction, perhaps with a figure. Details of how the software works or is used should be in the appendices.

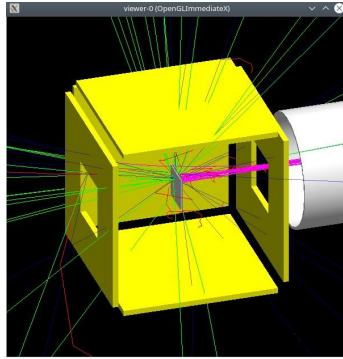


Figure 6: GUI Visualisation of muon beam interacting with the sample and the sample plate, using the `vis.mac` file and code in `musrSim`

One of the standard tasks in experimental physics is the comparison of measurements to theoretical models. These models can be simply described as mathematical expressions (functions) providing predictions of measured data. Visualisation of the data is quintessential to achieve this objective, besides the necessity of implementation of corrections & parameter transformations. Models based on quantum theory often predict the probability density function (PDF) of measurements dependent on a set of parameters. The data volume associated with these comparative studies is large. As a result, a computationally stable and feature-rich framework like ROOT is needed.

ROOT is a data analysis and I/O software framework developed as an open-source project coordinated by CERN, Geneva. Some of the prominent features associated with it are a GUI for interactive data analysis and a C++ programming interpreter. ROOT also offers compatibility with Python with its PyROOT functionality, allowing the user to analyse their μ SR data on multiple platforms for drawing sound conclusions to their scientific endeavours. It serves as the other half of the important software needed to run `musrSim` & `musrSimAna`, which have been covered in the section below in depth.

`musrSim` & `musrSimAna`

`musrSim` is a program based on Geant 4 platform, optimised to simulate the response of μ SR instruments to muons and their decay particles (typically electrons & positrons) during muon experiments. The software was developed by a group of scientists from Paul Scherrer Institut (PSI) and University of Oxford using the Geant4 and ROOT libraries. The particles' interaction with the instrument environment and detectors are simulated using the Monte Carlo algorithm.

It is essential that the user familiarises themselves with the `musrSim` "macro" (.mac) file as it will help them in performing basic μ SR simulations and data analysis, whilst laying a strong foundation for advanced μ SR simulations and data analysis.

For basic simulations, the following are the critical sections for users in the instrument definition "macro" file:

1. Degradation foil, sample & sample holder definition
2. Magnetic Field definition
3. Visualisation (enable/disable 3D GUI window with the help of a special macro file "vis")
4. Particle gun definition

musrSimAna is another program which is meant to help the user to analyse the output of the *musrSim* simulation run, generated in the form of a ROOT-compatible file. The output file contains a variety of graphical output and histograms that the user can analyse using the ROOT platform. The user can visualise the data from the simulation run associated with the correct *musrSim* "macro" (.mac) file from the ROOT-compatible .v1190 file generated using *musrSimAna*.

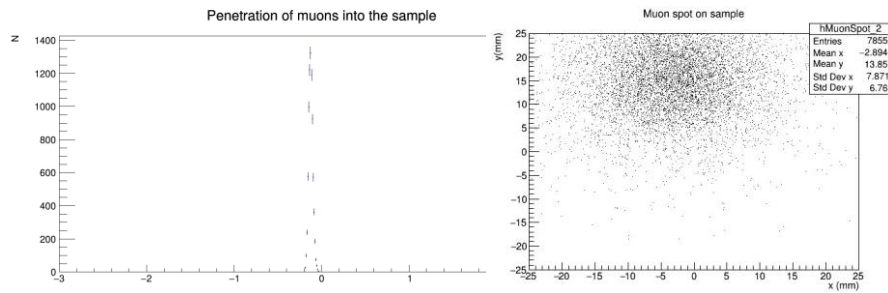


Figure 7: Muon implantation along sample depth (z-axis) and cross-section (xy-plane)

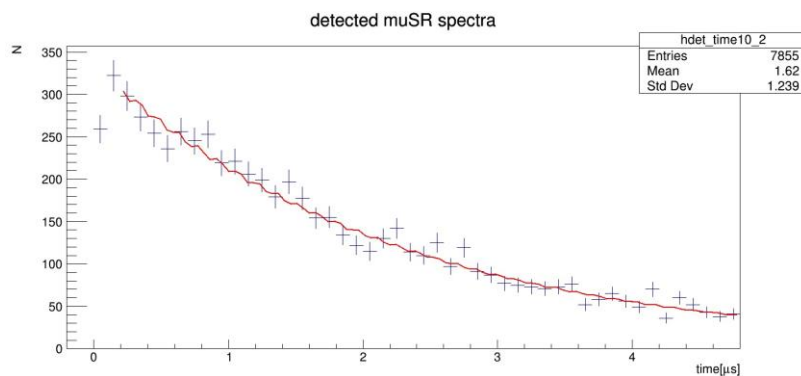


Figure 8: Predicted μ SR signal obtained from a sample simulation.

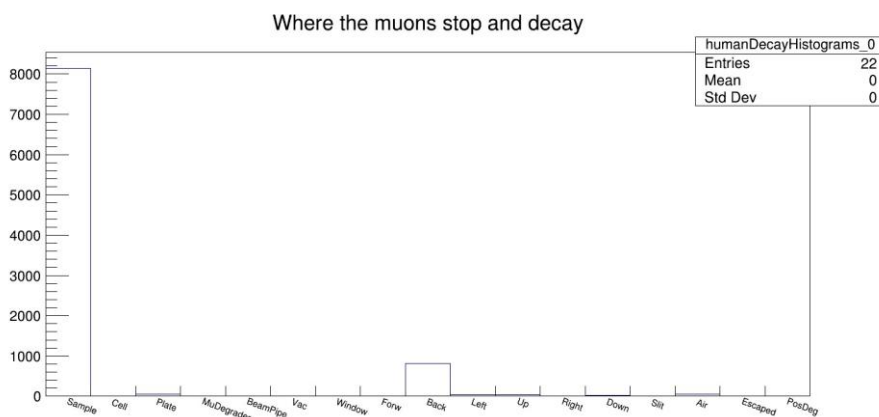


Figure 9: Histogram depicting where muons stop and decay along the beamline.

As seen from Figure x, the Root tree presents ‘raw’ analysis on the data, such as where the muons stop and decay, the energy spectrum and some additional information that may be defined in the scope of the macro file. However, other complex studies like studying asymmetry and counting rate, analysing the μ SR signal amplitude, time-dependent background can be examined using *musrSimAna*.

The advantages and disadvantages of using *musrSim* and *musrSimAna* can be listed in Table 1:

Advantages	Disadvantages
Ease of editing the parametric definition file ("macro" or .mac file), simple way of defining instrument geometry, muon beam and other features.	The user is required to have prior installation of Geant4 and ROOT before they proceed to install and execute <i>musrSim</i> and <i>musrSimAna</i> .
Potential to use the program for calculating muon stopping profile or in development of detector components.	The simulation of one event takes much more time than the measurement of a real event, forcing the user to access appropriate complex computational resources.

Table 1: Advantages and disadvantages of *musrSim* and *musrSimAna*

The default units of *musrSim* in input as well as output files (.mac and .v1190 files respectively) can be stated as follows:

Quantity	Default Unit
Length	mm
Time	μ s
Energy	MeV
Momentum	MeV/c
Magnetic field	T (tesla)
Electric field	keV/mm
Angle	degree

Table 2: *MuSRSim* default units for given physical properties.

SRIM & TRIM

Since the 1960s, methods to calculate the motion of ions and energised particles in a medium have been developed for computational simulations. These methods are the dominant way of theoretical treatment of stopping range problems. The underlying idea behind these methods is the concept of following the ion’s movement in the target by simulating the collisions with the atomic nuclei in the target. The electronic stopping power is usually considered as a frictional force slowing down the ion, as per the Ziegler-Biersack-Littmark (ZBL) model for stopping power which has been expressed in the equation below,

$$S(E) = S_e(E) + S_n(E)$$

where $S(E)$ is the cumulative stopping power through a given medium/target for a given beam energy E , $S_e(E)$ is the electronic stopping power of the target (a result of the inelastic collisions of the ion/particle with bound electrons in the sample), and $S_n(E)$ is the nuclear stopping power of the target (a result of the elastic collisions of the projected ion with the atoms and their nuclei in the sample). It must be noted that radiative stopping power has been neglected as majority of muon beamlines make use of surface muons which have a low energy momentum with 100% spin polarisation and a narrow energy distribution/spread.

Conventional methods for ion stopping range calculations are based on the BCA (binary collision approximation) , as stated by Robinson et.al. In these methods the movement of projected ions in

the implanted sample is treated as a succession of individual collisions between the recoil ion and atoms in the sample. For each individual collision the classical scattering integral is solved by numerical integration, given by the stated formulation.

$$\sigma = \int \frac{d\sigma}{d\Omega} d\Omega$$

where $d\sigma/d\Omega$ is the differential cross section, defined as a ratio of the number of particles scattered in a specified direction per unit time per unit solid angle N to the incident flux j . In classical scattering theory, the angle of scattering is determined by the impact parameter b . For a given incident flux, the number of particles scattered into solid angle $d\Omega$ can be expressed as:

$$\frac{d\sigma}{d\Omega} = \frac{N}{j} = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right|$$

In these methods the impact parameter is determined either from a stochastic distribution using the Monte Carlo (MC) algorithm, or in a way that considers the crystal structure of the sample. The algorithm provides the user with a distribution on the likelihood of results through random sampling, where the solution's accuracy is proportional to the number of samples taken. It must be noted that unlike conventional computational methods, MC analytic solutions do not converge, making it an ideal tool for simulations targeting design optimisation.

The recommended program for our case is **TRIM/SRIM**, which is based on the ZBL model. It has a user-friendly interface and has default parameters for all ions in all materials up to an ion energy of 1 GeV, which has made it immensely popular. However, it doesn't take account of the crystal structure, which severely limits its usefulness in many cases. Several BCA programs overcome this difficulty; one of them being **crystal-TRIM**, a variation of the original **TRIM** code.

BCA has successfully described many physical processes but with its own limitations, like providing a real description of the slowing down process of energetic ions. The basic assumption that collisions are binary results in severe problems when dealing with multiple interactions. Also, in crystalline materials, the selection process of the next colliding lattice atom and the impact parameter always involve several parameters that could affect the results. Reliability in BCA-based MC simulations is obtained by including multiple collisions in the calculations, which is not easy to do correctly.

SRIM is a software package developed by James Ziegler, M.D. Ziegler & J.P. Biersack, simulating and providing data about the Stopping and Range of Ions in Matter. **TRIM** (Transport of Ions in Matter) is a group of computer programs that calculates the interactions of ions with targets using the Monte Carlo approach. SRIM can provide the user with the much-needed data about the stopping range of ions alongside estimates of the stopping energy. It must be noted that for our purpose of simulating muon experiments, we would be modelling muons as heavy electrons. Henceforth, muons would be referred as *ions* when we are having a conversation about SRIM & TRIM. Note that such an assumption is not required when using *musrSim* & *musrSimAna*.

Using the data from SRIM, the user can visualise their experiment using TRIM. TRIM can accept inputs of complex targets made of compound materials up to 8 layers. It can also calculate the final 3D distribution of the ions and all kinetic phenomena associated with the ions. For majority of basic simulations that user would do, we will assume that the user is performing an "*Ion Distribution and Quick Calculation of Damage*" type of TRIM simulation using SRIM data. However, TRIM allows user to perform other calculations, should the user be interested.

The program assumes that there is cylindrical symmetry in the final ion distributions. The assumed cylindrical axis is perpendicular to the target surface at the point of ion impact. (If the initial ion beam is not normal to the target surface, then none of the following is valid.) The "radial"

Commented [SD(8)]: Review

Commented [BP(9R8)]: Use the information in the PDG explanation of Muon Stopping and Range to explain the general background. See if anything further is then needed for the specific of SRIM/TRIM.

parameters refers to these cylindrical coordinates. The "projected" parameters assumes that an X-Y plane has been inserted through the axis, and the final ion distribution is projected onto this plane. This projection is identical to the view on the screen and marked as the "XY Plane" (X is the depth axis; the XZ plane is perpendicular to the XY plane).

The mean "lateral range" of the ions is zero in both the radial and projected definitions if there is cylindrical symmetry. So the lateral range in TRIM is defined as the mean absolute value of the lateral range. The ion straggling has its normal definition as the second moment of the distribution. The lateral projected parameters directly describe the final distribution under a mask edge as used in VLSI technology. Remember, the total ion concentration directly at a mask edge is one half of the concentration of a uniform implantation distribution.

These words are names for quantities related to the second, third and fourth moments of the ion distribution. They are important because many programs require these parameters to create analytic functions of ion implantation distributions. The use of these words in the ion implantation field is different from that in some statistics textbooks. In this field, we use mostly the definitions first proposed by B. Winterbon ("Ion Implantation Range and Energy Distributions", vol. 2, Plenum Press, 1975). For convenience, we define each moment below in several different standard notations. It is assumed below that the ion trajectory begins perpendicular to the target surface, co-linear to the x-axis, and the y-axis and z-axis are orthogonal in the target surface plane (the TRIM 4-plot display shows trajectories projected on both the XY and the XZ planes).

$$\text{Mean Projected Range } R_x = \frac{\sum_i x_i}{N} = \bar{x}_i$$

$$\text{Lateral Projected Range } R_y = \frac{\sum_i |y_i|}{N} = \bar{y}_i$$

$$\text{Radial Range } R_r = \frac{\sum_i \sqrt{y_i^2 + z_i^2}}{N}$$

where x_i is the projected range of ion i on the x-axis, i.e. the perpendicular distance from the surface to the end of an ion's track; $\sum_i x_i$ = sum of the ion ranges; $\sum_i x_i / N$ = the mean projected range of N ions; and $\bar{x}_i$ = the mean projected range of all ions. The transverse coordinate y is treated the same, only the distance is taken in the XY plane. The mean projected lateral range is zero for a perpendicular beam, so the above lateral range definition averages the absolute values to provide other information on the first moment of the radial spread. The mean radial range assumes cylindrical symmetry of the ion distribution.

Variance is the second moment of the range distribution, and we show below several identical definitions using various common notations.

$$\text{Variance} = \frac{\sum_i (x_i - R_x)^2}{N}$$

Straggling is a word which is used in ion implantation in several ways, and care should be taken to determine the author's definition. Sometimes it is a synonym for variance and sometimes it is defined as the square root of the variance. In other cases, an author such as Winterbon, uses normalized definitions such as: Straggling = $\sigma^2 / \bar{x}^2$. We use the common definition that straggling is the square root of the variance:

$$\text{Straggling} = \sigma = \sqrt{\frac{\sum_i (x_i - R_x)^2}{N}}$$

We define lateral straggling in the same way as range straggling above. (Lateral coordinates are sometimes called Transverse coordinates.) For a normally incident beam we can assume cylindrical symmetry of the range distribution, so the mean lateral projected range is zero (i.e. $R_y = 0$).

One of the key assumptions made in using TRIM for simulations of muon experiments like μSR is that the muon is modelled as a light hydrogen ion with 0.113 times the mass of the normal hydrogen ion. This assumption holds true for positive muons. In order to model negative muons, they can be modelled as heavy electrons by making minor modifications to the TRIM code. The Bethe-Bloch equation serves as the underlying principle in order to understand the stopping range and power results from TRIM for the muons modelled as light ions at high velocity.

The two major assumptions associated with the stopping of high velocity light ions in matter are (a) the ion is much heavier than the target electrons, which is true for a muon and (b) the ion is moving much faster than the target electrons and is fully stripped of electrons. These two assumptions are satisfied by the modelling assumption of muons being treated as light ions. The beam energies for the Bethe-Bloch domain for muon stopping power lies between from 1 MeV to 10 GeV. The original Bethe-Bloch stopping formula S may be stated as:

$$S = \frac{4\pi e^4}{m_e v^2} Z_t^2 \left[\ln \frac{2mv^2}{\bar{I}} - \ln(1 - \beta^2) - \beta^2 + \Psi(Z_t) \right]$$

Where $\bar{I}$ is the averaged excitation potential per electron and $\Psi(Z_t)$ is the Bloch error which can be corrected by introducing the *Shell Correction* and *Density Effect* term.

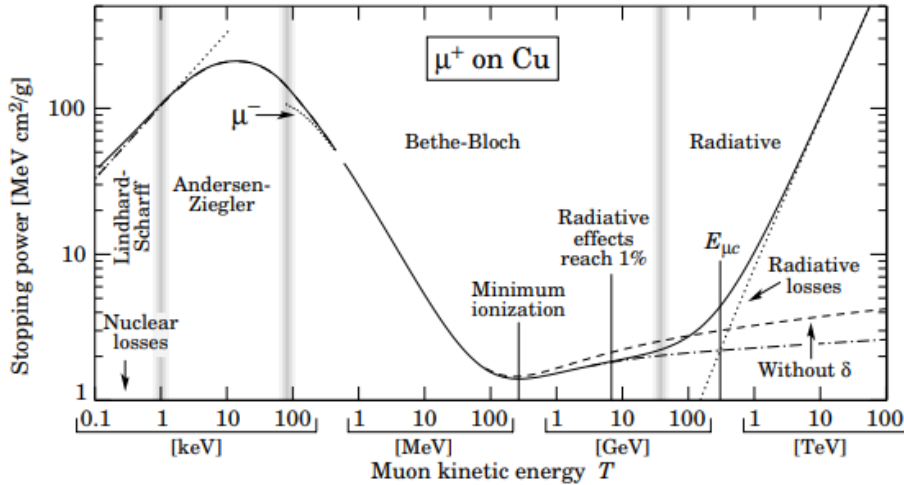


Figure 10: Stopping Power vs Muon beam energy domain curve

Muon Experiment Decision & Optimisation Process (MEDOPS)

Literature Review & Solution-Neutral Problem Statement

A key consideration for any μ SR experiment is whether the measured muons are in a frequency range of interest. Hillier et al. have stated that a pulsed muon source should be the optimal choice when many muon lifetimes need to be measured at low frequency pulsed stimuli, while a continuous muon source should be used for high frequency and fast relaxation time measurements.

The mass and size of a sample are important to consider during sample mounting because a situation where we have a small sample, and a big sample holder can be problematic. This is where a competition between areal mass density, beam spot size and experimental statistics plays out. Areal mass density and sample area are the key critical design variables for a muon spectroscopy experiment the user must consider, from a sample's perspective. The user must also consider the beam cross-section (spot size) and beam momentum, as it will determine how many muons are implanted inside the sample. The optimum compromise between muon beam spot size and sample design variables can be done by 2 methods: (a) controlling the spot size by adjusting the slit positions, and (b) optimising the geometry of the sample.

Ideally, all muons should stop at a mean depth inside the sample. However, this does not turn out to be the case due to the phenomenon of straggling. Both stopping power and range should be understood as mean values over a group of atoms or molecules, because energy loss is a statistical phenomenon. The fluctuations observed in these statistics is referred to as straggling, and the most important among them is lateral range straggling, where particles in the same medium have varying path lengths between the same initial and final energies. SRIM has modelled range and lateral straggling as functions of variance of the range distribution. The following are their respective mathematical definitions:

$$\sigma_{xx} = \left[\frac{\sum_i x_i^2}{N} - R_p^2 \right]^{\frac{1}{2}} = \langle (\Delta x_i)^2 \rangle^{\frac{1}{2}}$$
$$\sigma_{yz} = \left[\frac{\sum_i \left(\frac{|y_i| + |z_i|}{2} \right)^2}{N} - R_p^2 \right]^{\frac{1}{2}}$$

Where $\Delta x_i = x_i - R_p$, with x_i being the projected range of the muon particle i on the x -axis and R_p being the mean projected range which is calculated as $\sum_i x_i / N = \langle x \rangle$

Research done by Taller et.al and Greene et.al has shown that all the above critical design variables has an effect on the degrader foil design. It is obvious that degrader foils causes a loss in energy of the particle, which results in a shallower penetration depth into the target. Depending on the changes made to areal mass density and sample area, degrader design can be affected by the required implantation depth.

Commented [BP(10)]: Explain straggling in more detail. Might be worth introducing in the introduction.

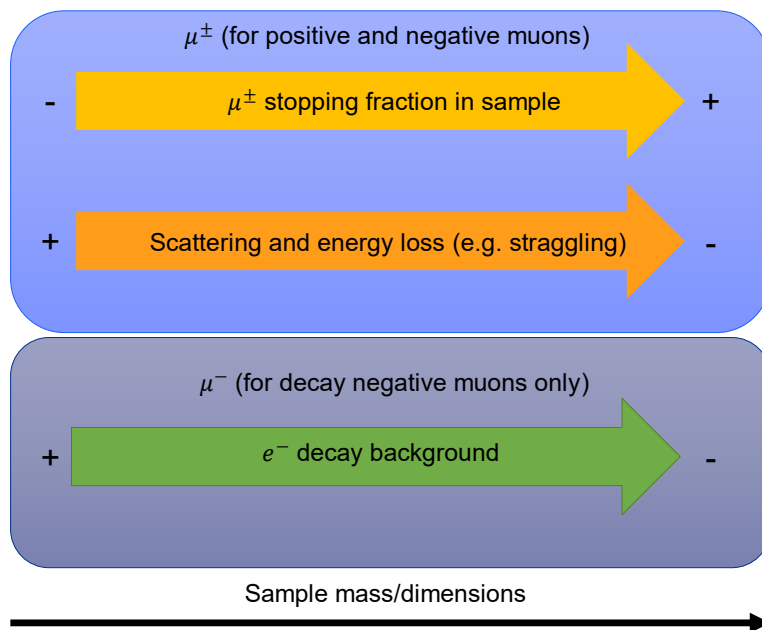


Figure 11: Different trends to be considered for sample optimisation while planning μ SR experiment.

The figure x, originating from Rohrken et al. with modifications, is an illustration of important first order effects for optimisation of muon stopping target for the Mu2e experiment, with dependency on the target mass. The illustration is quite relevant because the same effects have been observed when users were considering sample design in previous μ SR experiments. The illustration can be interpreted in the following manner:

1. **Target mass v/s number of stopped muons:** The mass of the stopping target (in our case, the sample for the μ SR experiment) would determine the range the low momentum muons of the beam need to traverse before coming to a stop. The higher the mass of the sample, the more muons will come to rest within the physical volume. Sometimes, increasing the sample mass may have adverse effects for a given experiment. Increasing the number of stopped muons can effectively increase the amount of material need to be traversed by the positron/electron (outcome of the conversion process) to reach the tracking detectors.
2. **Target mass v/s scattering and energy loss:** More sample can potentially increase the effects of multiple Coulomb scattering and energy loss caused due to phenomena like straggling of outgoing positrons. As a result, there is a noticeable decrease in the particle momentum and the electromagnetic character.
3. **Target mass v/s background:** The target mass directly determine the scale of the dominant background – e^+ or e^- that originate from the weak decay of bound muons decaying in orbit of the target nuclei. Further first and higher order effects are the matter distribution and geometry of the sample, the position of the sample relative to the detectors and within the magnetic field. This results in other irreducible background sources, for example electrons originating from radiative pion captures.

Commented [BP(11)]: Explain the considerations relevant to muon spectroscopy (1 and, a little bit, 2) here. Remove the diagram above.

The DSO study for samples used in μ SR experiments follow 2 main objectives – baseline design of the muon stopping target, and alternative stopping target materials & configurations. Rohrken et al. has performed simulations with variations in the number of foils, foil thickness/radii, spacing and location within the magnetic field suggesting setups with thinner multiple foils made of materials having a large atomic number Z , resulting in improved efficiency of signal reconstruction and reduction in expected background.

Before the experiment starts, sample mounting is an important factor needed to be taken into consideration, as it can be the difference between experimental success and failure. Generally, a sample holder must be used to reduce the background term and to minimise relaxation of polarised muons in sample. Large sample cells exist for gaseous samples, where a large amount of gas is needed to stop the muons. However, different sample holders and sample environment may have their individual muon relaxation rate which might be temperature dependent. Other practical constraints that may be associated with the optimisation problem are factors like the distribution of a powdered sample within the sample holder.

From the literature review presented in the previous section, we can define the solution-neutral problem statement.

“Optimise the sample dimensions presented to the muon beam, subject to maintaining enough quantity to stop all of the incident muons in the sample.”

Decision Flowchart

It must be noted that for basic simulations and majority of muon experiments, the following are the key variables that users should be aware: **the muon beam features, sample mass and sample area**. These determine how many muons are implanted in the sample compared to those on the surface, when the sample is place in a muon beam with a particular **spot size** and **beam momentum**.

The methodology and decision flowchart in Figure x shows that the multi-objective multivariable optimisation problem is solved systematically and, in more complex cases, using Monte Carlo (MC) simulations. This approach is easily applied to basic experiments, but also for experiments of greater complexity with more variables, e.g. pressure cells and in-operando experiments. The flowchart has been setup in a manner such that it could be used regardless of the nature of the input parameters, thus accounting for cases where optimisation may not be necessary.

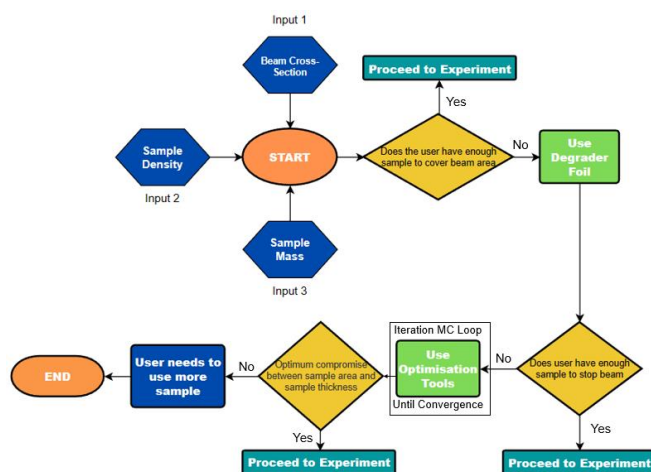


Figure 12: User Decision Flowchart for a μ SR experiment

Commented [BP(12)]: I think that is for a specific type of sample plate (Fe2O3) rather than a general statement. This sentence certainly needs to be changed to be generally correct.

Commented [SD(13R12)]: I believe we can do with that statement in entirety.

Commented [BP(14)]: There are some practical constraints too.

Commented [SD(15R14)]: Do I need to elaborate on it in further details?

Commented [BP(16)]: Needs a statement about prior art and also how this can be used simply without simulations.

Commented [BP(17R16)]: Where does the relevant information come from for the first two steps?

An example where this methodology & decision flowchart could be used is the sample & degrader design optimisation for a basic μ SR experiment conducted using EMU. In this case, the process will follow accordingly:

1. User input – the mass & density of the sample, instrument input – EMU beam cross section (approx. 5 cm^2). It must be noted that besides the beam cross-section, the beam spot shape is significant as well because it can often determine the quality of the signal received and the muon relaxing asymmetry.
2. Check if user has enough sample to stop muon beam / cover beam area without degraders (sample areal density $\geq 180 \text{ mg/cm}^2$). If yes, proceed to experiment. If no, use degrader foil and proceed to step 3.
3. Check if user has enough sample to stop muon beam / cover beam area without degraders (sample areal density $< 180 \text{ mg/cm}^2$ and $\geq 70 \text{ mg/cm}^2$). If yes, proceed to experiment. If no, use optimisation tools until convergent solution is obtained and proceed to step 4.
4. Check if optimum compromise has been achieved between sample area & sample thickness post-optimisation. If yes, proceed to experiment. If no, user concludes that they need to use more sample before proceeding with their experiment.

Commented [BP(18)]: Relevance of beam shape

There could be more inputs & variables depending on the complexity of experiment done by the user. Users might wonder where the MESSC fits within the larger picture. It can be declared that the underlying design behind the software caters to all stages of the decision flowchart, especially as the "set of optimisation tools" if the user concludes that they do not have enough sample to cover the beam spot and to stop the muons (with or without degrader foils.)

Process Methodology

The flowchart below is a simple illustration of a typical computational simulation of a μ SR experiment using MESSC, with description of the involved steps.

Commented [BP(19)]: Make the flowchart a figure then it can be referred to elsewhere more easily.

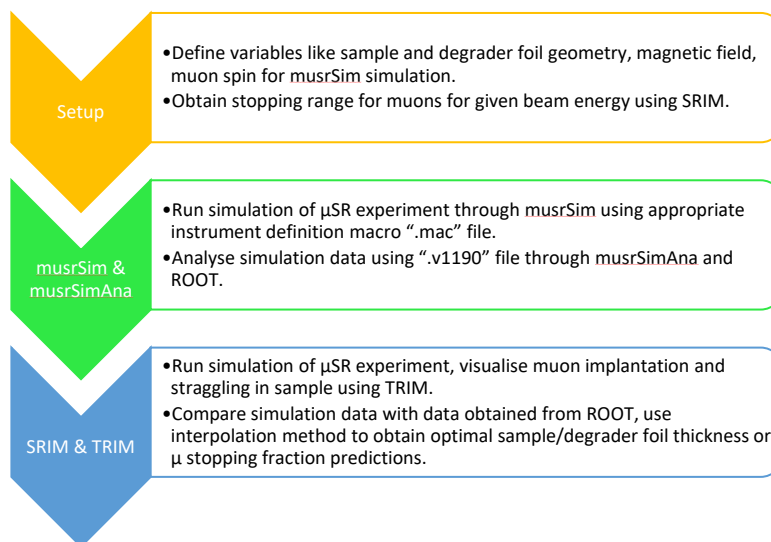


Figure 13: MEDOPS Optimisation methodology

Using the flowchart shown in Page y, an ISIS user performing μ SR experiments can generate computational data which they can compare with their real-time experimental data. The process flowchart can help a user in validation of their work. However, it is the recommendation that the user uses the process flowchart when they are in the first 2 stages of the decision flowchart shown in Figure x of the *Decision Flowchart* section.

If the user is unable to satisfy the conditions as stated in the first 2 stages of the decision flowchart, the situation at hand becomes a multi-objective, multivariable optimisation problem which can be solved using the Monte Carlo approach. The process flowchart shown above can be adapted such that the process is iterative in nature (unlike linear for majority of cases) and follows the Monte Carlo algorithm, demonstrated by Figure x. The simulation process is computationally intensive, but it can provide the user with the optimal compromise between sample area and sample depth such that the user can proceed with their experiment with or without degrader foils. It must be noted that during the process, the solution does not reach convergence unlike other optimisation techniques.

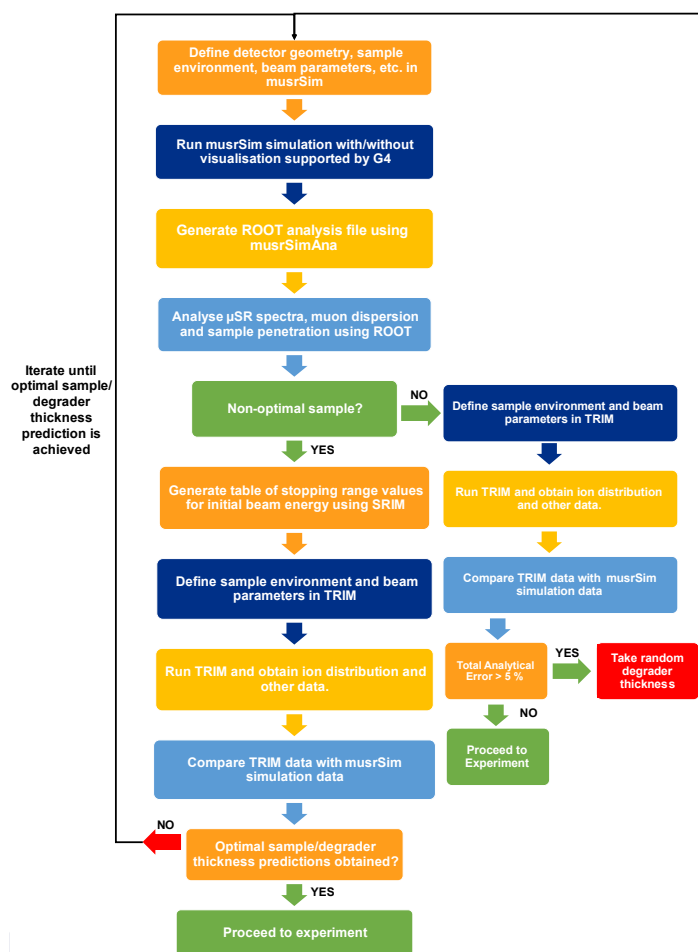


Figure 14: Iterative Multiobjective Monte Carlo Optimisation Loop

There could be other variables that might affect setup variables other than the decision flowchart. One of the primary variables needed to be considered is the type of study the user is conducting dependent on the μ SR experiment. In such cases, it is obvious that the setup for the *muSRSim* and SRIM would adapt and change accordingly. For example, setup for negative muons-based experiments (like elemental analysis) differs from the standard setup for experiments using positive muons like studies on semiconductors and superconductivity. Nevertheless, there are no other significant changes that make the flowchart and the optimisation solution incompatible for certain set of studies and conditions.

1. SETUP (Use MuSR): Define materials, geometry, physics process, particles and other variables. MuSR is a derivative of Geant4. Setup type of MuSR simulation to be performed.
2. RAW DATA: MuSRSim Simulation, ".v1190" file generation, run MuSRSimAna, ".root" file generation
3. SRIM Setup: The type of experiment (see Step 2) will determine SRIM table setup. For example, experiments performing study on organic chemistry/biosystems will have a different setup compared to experiments that are going to use gas cells. Experiments making use of solid elements or compounds have a standard procedure.
4. TRIM Setup: The type of experiment (see Step 2) will determine TRIM simulation and data. Experiments focussing on electronic irradiation, elemental analysis, carbon have special set of instructions as compared to standard "ion implantation" simulation.
5. SRIM data will give stopping range of ions for initial beam energy. TRIM data will give NEP (neutron, electrons and photons) transmission file.
6. Estimate the depth at which protons will implant themselves into the sample in the final beam energy using interpolation.
7. ROOT analysis for data obtained from ".root" file generated by MuSRSim & MuSRSimAna
8. Generate table for stopping energy of protons in target material using SRIM, obtain final target thickness through difference between desired and final beam energy.
9. Repeat Steps 4 – 10 until desired thickness is achieved.
10. Manufacturing, instrument configuration, spectral results, validation of model and study.

Process Verification & Validation – Sample Design & Muon Implantation Study

Experimental Outcomes, Planning and Setup

To verify the simulation approach experimentally, measurements were taken on different packaged hematite powder with varying degrader thicknesses. The MEDOPS & MESSC Verification & Validation (M2-V2) Experiment aimed to perform verification and validation (V&V) of the MEDOPS by performing real-time experiments for common scenarios of users and comparing the results with computational simulation data provided by MESSC. This process will help in demonstrating that the MEDOPS and MESSC are quintessential towards obtaining a good signal from the μ SR instrument by applying the two entities in a simulated μ SR experiment, thus resulting in process and operational qualification (PQ/OQ), subject to successful runs.

The predicted outcomes at the beginning of this study can be listed as follows:

1. Verification & validation of MEDOPS & MESSC by implementing the process during the calibration process leading to PQ and OQ,
2. Quantification of benefits of MEDOPS & MESSC implementation for users by measuring attributes like *reproducibility, reliability, repeatability, accuracy and precision* & identification of key optimisation parameters.

Commented [BP(20)]: To verify the simulation approach experimentally, measurements were taken on different packaged hematite powder with varying degrader thicknesses.

3. Identification of future work required on extracting *full potential* of MEDOPS & MESSC post-implementation in standard operating procedures,
4. Generating *visual results & data* for users to inspire confidence to use μ SR for R&D projects,
5. Lay foundation towards adapting MEDOPS for neutron experiments.

The decision was taken to proceed with the following setup:

1. Sample: Powdered sample of hematite (Fe_2O_3) with varying dimensions and constant mass of 300 mg
2. Sample Holder: 25 μm silver (Ag) foil packet to store the powdered sample.
3. Degradar foils: 25 μm titanium (Ti) foils
4. Sample plate: 50 x 50 mm silver (Ag) plate
5. Sample Environment: 298 K, 1 atm (s.t.p. conditions)

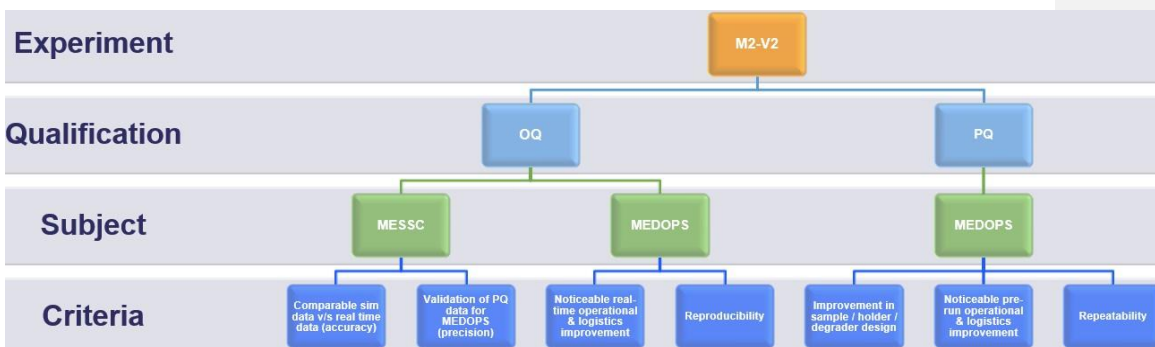


Figure 15: M2-V2 experiment qualification roadmap

Figure x is an illustration that led the experiment team to generate the necessary general **qualification matrix** prior to the experiment run. Each qualification had a separate set of criteria depending on the subject. For example, it is critical that the MESSC can generate simulation data which is comparable with the data obtained from the real-time experiment scenarios from operational perspective, which would also help in validation of the MEDOPS' process qualification. The team proposed five scenarios with fifteen minutes of allocated beam time to gather appropriate real-time computational data for further analysis & perform necessary studies needed for a successful V&V. The scenarios have been listed as follows:

1. SC-001: Only sample plate
2. SC-002: Random hematite sample with sample holder, sample plate with no degraders.
3. SC-003: Sub-optimal hematite sample with sample holder, sample plate and no degraders
4. SC-004: Sub-optimal hematite sample with sample holder, sample plate and degraders
5. SC-005: Optimised hematite sample with sample holder, sample plate and degraders

The team decided to simulate scenario SC-004 with three different setups – optimal number of degrader and +/- 3 degraders from the optimal number to demonstrate the effect on the signal by having too many or too less degraders for a μ SR experiment using a hematite sample with a sub-optimal design. From simulations done using the tools mentioned in MESSC (SRIM, TRIM and musrSim), the optimal experimental setup for the scenario SC-005 can be described as follows: Mylar (75 – 100 μm thickness) Ti degrader (8 x 25 μm) + Ag foil packet (25 μm) + hematite sample (150 μm) + Ag plate (50 μm).

Commented [BP(21)]: Cite a textbook for this.

It was decided that the MuSR instrument in the European Muon Facility at ISIS was going to be used for the collection of the experimental data. Under the supervision of Peter J Baker, the experiment was performed, with the following setup dimensions:

1. SC-001: 50 x 50 x 0.5 mm Ag Plate
2. SC-002: Hematite sample dimensions – 10.69 x 19.69 x 0.5 mm
3. SC-003: Hematite sample dimensions – 17 x 17 x 0.2 mm
4. SC-004: Hematite sample dimensions – 24 x 24 x 0.1 mm
5. SC-005: Hematite sample dimensions – 22.5 x 22.5 x 0.15 mm

The sample dimensions were conceived in a manner such that the Ag foil packet contained approximately 300 mg of hematite with negligible deviations in the quantity. Each measurement was meant to last approximately 10 to 15 minutes in duration, with specific interest in measuring the relaxing asymmetry of the muons in the setup and the resultant stopping fraction within the sample.

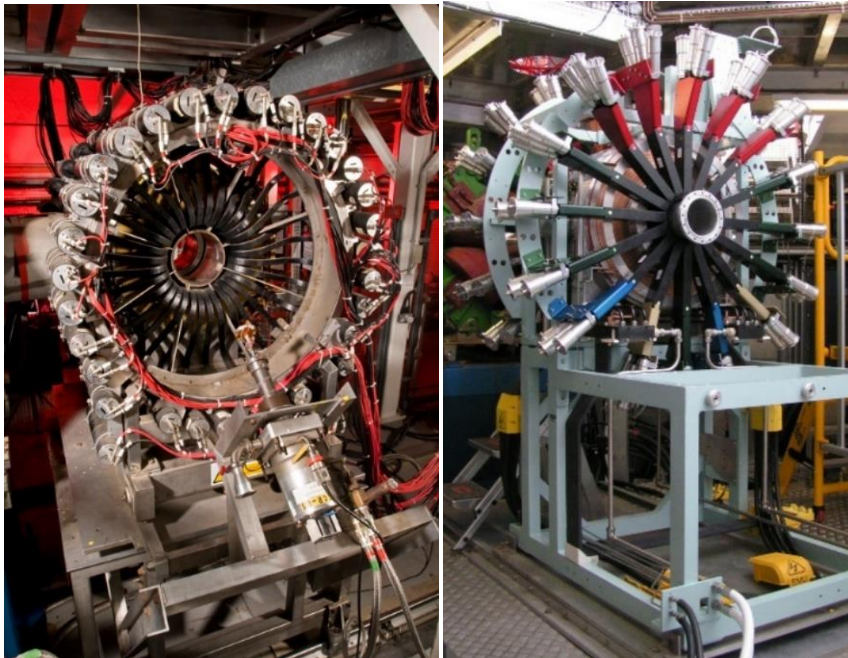


Figure 16: Current setup of two ISIS μ SR instruments – MuSR and EMU. It must be noted that the experiment is not dependent on the instrument.

Commented [BP(22)]: Need to establish what the measured quantity was and the expected error in that quantity given the counting statistics.

Commented [BP(23)]: I don't think this is needed as the instrument is essentially generic to the simulation.

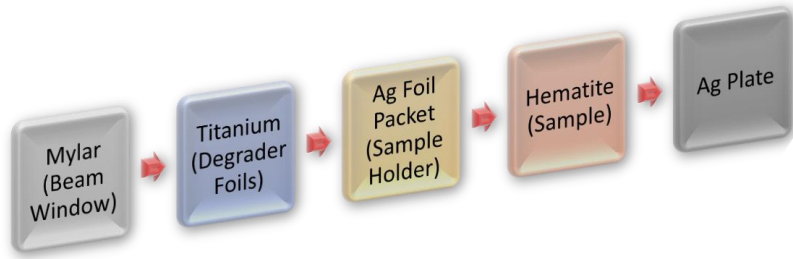


Figure 17: Layout of the sample workbench for the M2-V2 experiment, with the assumption of homogeneous mass distribution for the specified layers.

Results and Data Analysis

The data gathered from the M2-V2 experiment underwent comparative study with the Pareto multi-objective optimisation model to validate and verify the MEDOPS optimisation process methodology. We also compared the data with theoretical results provided by the Ziegler-Biersack (also TRIM) model and results from the Elson-Månsson model for TRIM simulations. Månsson et.al have compared their model data with μ SR experimental data collected from the GPD instrument at PSI.

For comparisons associated with the computational models, we were required to simulate the experimental setup for varying beam momentum/energies using SRIM/TRIM to obtain essential data that would allow comparative study with a new computational model proposed by us. This new computational model is supported by the MESSC framework, which means validation of our model will result in the successful verification and validation of the MESSC.

The Elson-Månsson model uses a Python-based GUI software which runs TRIM calculations in the background and provides the users with valuable results on muon stopping fractions. One of the advantages of the software when compared to standard TRIM calculations is they assume a Gaussian distribution of the beam momenta and energies. The software is also capable of collimating the beam area while accounting for random small angle divergence for each individual muon. The model requires 20 momenta upto 3σ (where σ is the standard deviation/spread of the distribution) away from the input mean momentum of the beam to obtain the desired energy distribution to predict the depth where majority of the muons would stop in the sample. The software then generates a list of energy values part of the beam using the following mass-energy relation expression:

$$E_i = \sqrt{(p_i c)^2 + (m_0 c^2)^2} - m_0 c^2$$

where, E_i is the value for beam energy corresponding to the momentum p_i , c is the speed of light in vacuum, and m_0 is the rest mass of the muons equalling to $105.658 \text{ MeV}/c^2$.

Commented [BP(24)]: Needs to be explained or a reference cited.

Commented [BP(25)]: These need more explanation and introduction somewhere.

The results for the Ziegler-Biersack model can be obtained by running a standard TRIM simulation for a specified beam energy. Often, the user would be provided information about the beam, which would involve information about the beam momentum and the spread of the momentum distribution. The user can then compute the corresponding beam energy value from the above mass-energy relation expression.

In our model, we initially performed three standard TRIM simulations for a specified beam energy calculated from the beam momentum data for a given instrument provided to us by the facility. The three simulations are setup in such a manner that the beam energies are equally spaced at an interval of 0.1 MeV at either ends. An example would be as follows: [3.70, 3.80 (specified beam energy), 3.90] MeV. To obtain the final range/energy distribution, we use the convolution theory in statistics which gives us with the following formulations:

$$\begin{aligned}\bar{\mu}_x &= (\mu_{x_1} + \mu_{x_2})/2 \\ \sigma_{\mu_x} &= \sqrt{\sigma_{x_1}^2 + \sigma_{x_2}^2} \\ \bar{A} &= (A_1 + A_2)/2\end{aligned}$$

The final distribution obtained is the convolution of the two distribution functions on the either end of the beam energy. It has the following expression:

$$y = \bar{A}f(\bar{\mu}_x, \sigma_{\mu_x})$$

where $\bar{A}$ is the mean amplitude, $\bar{\mu}_x$ is the mean energy/target depth/beam momentum/areal density, σ_{μ_x} is the standard deviation and f is a probability density function.

The density function f takes the form of a normal distribution function (also referred to as Gaussian distribution function) if it is confirmed from comparing either *musrSim* or TRIM results with the beam camera image that the nature of the beam is Gaussian in nature. In such cases, the mathematical expression can be re-written as follows:

$$y = \bar{A}e^{-\frac{1}{2}\left(\frac{x-\bar{\mu}_x}{\sigma_{\mu_x}}\right)^2}$$

The model is accurate at predicting the number of muons stopping at an optimal target depth but has shown approximately 10% difference in the spread of the distribution (σ) in comparison to the Elson-Månsson model. This was due to the model not accounting for straggling. As per the Ziegler model, straggling is the second moment (standard deviation/ σ) of the muon distribution curve for specified beam energy. Thus, the formulation for the corrected standard deviation in the original Baker-Sengupta model can be expressed as the following:

$$\sigma_{\mu_x} = \sqrt{\sigma_{x_1}^2 + \sigma_{x_2}^2 + \sigma_{straggling}}$$

where the general expression for $\sigma_{straggling}$ can be given as follows:

$$\sigma_{straggling} = \sqrt{\left(\frac{\sum_{i=1}^n \sigma_{xx_i}}{n}\right)^2} = \overline{\sigma_{xx}}$$

where n is referred to the number of simulations at individual beam energies performed by the user.

It is the recommendation that the number n be somewhere between 2 to 6, with beam energies equally distant from the mean beam energy. An example has been illustrated in Figure x, using code snippet written in Python kernel using the Jupyter Notebook platform.

```
In [29]: print(np.linspace(3.6, 4.0, 5))

[3.6 3.7 3.8 3.9 4. ]
```

Figure 18: Range of beam energies to be simulated by user with mean energy at 3.8 MeV and number of simulations $n = 5$

If the longitudinal straggling correction is implemented in the original model, the corrected/optimised model (referred to as Optimised Baker-Sengupta model) gives us an optimal solution with a difference ranging from approximately 1% to 2% in the distribution spread when compared with the Elson-Månsson model while retaining the accurate predictions made for target depth and number of muons stopping inside a specified target from the original Baker-Sengupta model.

Commented [BP(26)]: This needs more explanation.

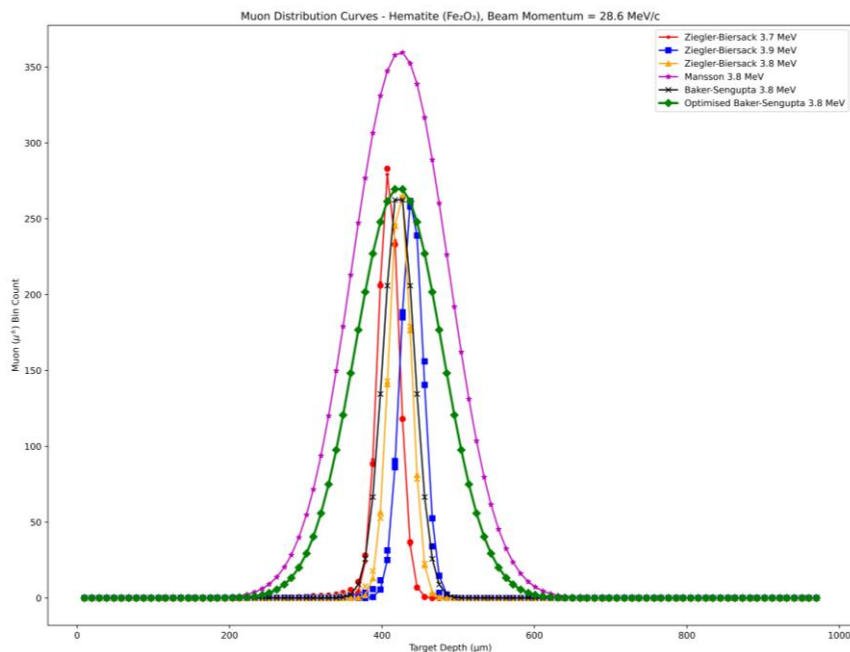


Figure 19: Muon distribution curves with respect to target depth for powdered hematite sample for a specified beam momentum using different models.

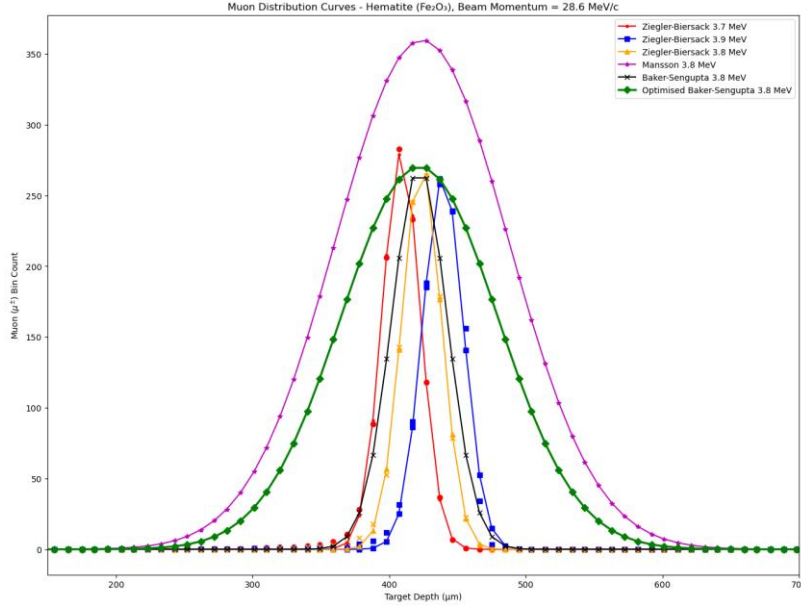


Figure 20: Muon distribution curves with truncated x-axis.

Using the MESSC framework and Python, the range curve for the different models was derived from the muon distribution data. The range curve can be plotted as a function of areal density by referring to the following formulations:

$$\begin{aligned}\widetilde{\Sigma\mu_i^\pm} &= \widetilde{\mu_i^\pm} + \widetilde{\Sigma\mu_{i-1}^\pm} \\ \rho_{A,setup} &= \rho_{setup} \cdot x\end{aligned}$$

From the above expressions, we can summarise that the range curve is the cumulative distribution function (CDF) of the probability density function $y = \bar{A}f(\bar{\mu}_x, \sigma_{\mu_x})$. The cumulative muon number can be expressed in terms of the muon stopping fraction. In the case of a Gaussian, the range curve obtained from a model will be the CDF of the Gaussian muon distribution function with respect to the target depth or areal density, as shown by Figures x. The density of a given target can be calculated using the following expression,

$$\rho_{setup} = \sum_{i=1}^k \rho_i \chi_i$$

where, k is the number of layers of material in the target, i being the index number of the material layer starting from layer 1, ρ_i being the density of the material of the target layer, and χ_i being the mole fraction which can be expressed as a ratio of layer thickness to the overall target thickness.

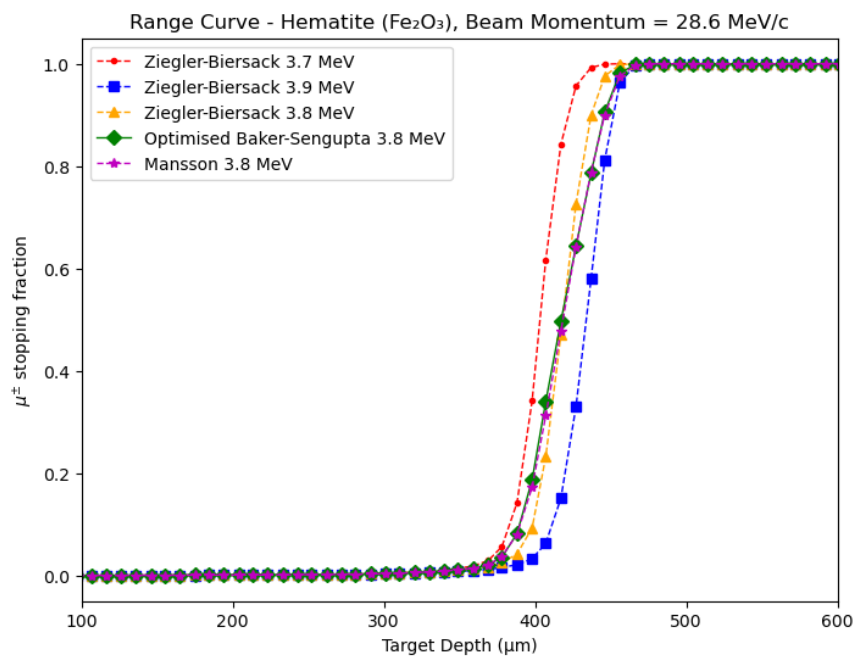


Figure 21: Range curve plotted as a function of target depth x

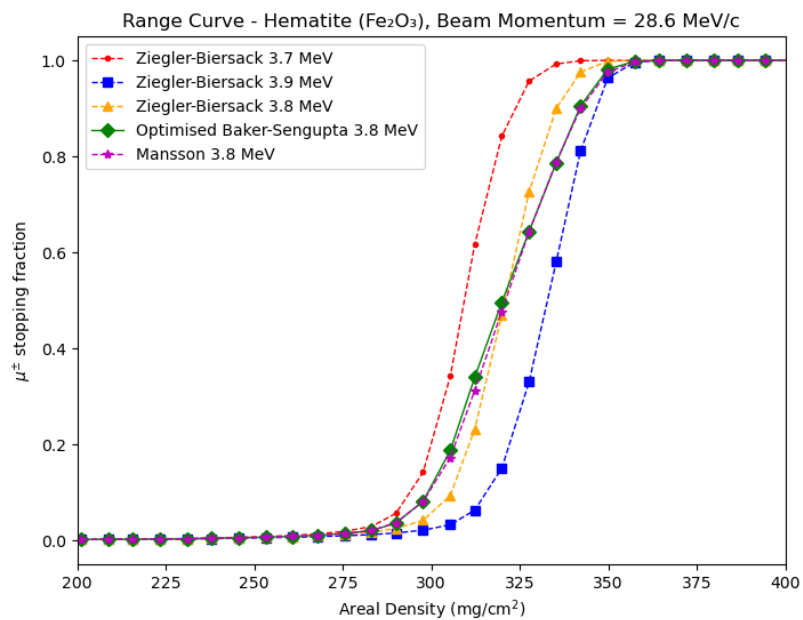


Figure 22: Range curve plotted as a function of areal density with the units of mg/cm^2 .

The models were also tested against the experiment conducted at PSI by Månsson et. al for further verification. The computational model accounted for the setup of the GPD instrument, the pressure cell made out of MP35N (Nickel-35, Cobalt-35, Chromium-20, Molybdenum-10), the Variox-Heliox cryostat and the chromium chloride (CrCl_3) sample. The mean beam momentum was 100 MeV/c (approximately 40 MeV of energy), which meant the model was being tested for μSR experiments done in decay muon beamlines, unlike surface muon beamlines in the previous case. For the TRIM simulations, the geometry was setup in the following format: Mylar (0.1 mm thickness) + Aluminium Nitrogen (0.2 mm) + Copper (0.8 mm) + Brass (2 mm) + MP35N (front of pressure cell, 9 mm) + CrCl_3 (6 mm) + MP35N (back of pressure cell, 9 mm). In this case, we assumed that the sample occupied the entire sample cell space in the pressure cell.

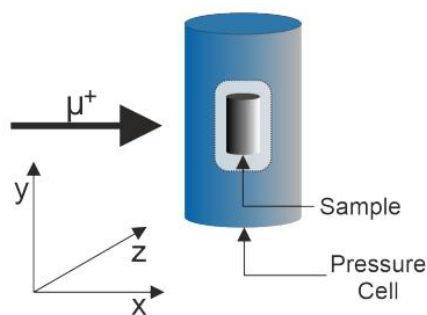


Figure 23: Geometric setup in the simulation, arrow indicating direction of muon beam momentum.

The results generated showed similar trends to the one observed during the analysis of results from the M2-V2 experiment, albeit showing a major difference of approximately 30 – 40% in the spread between Månsson method and the original model without straggling correction, which indicates the importance of straggling correction grows increasingly when dealing with muon with energies greater than 10 MeV.

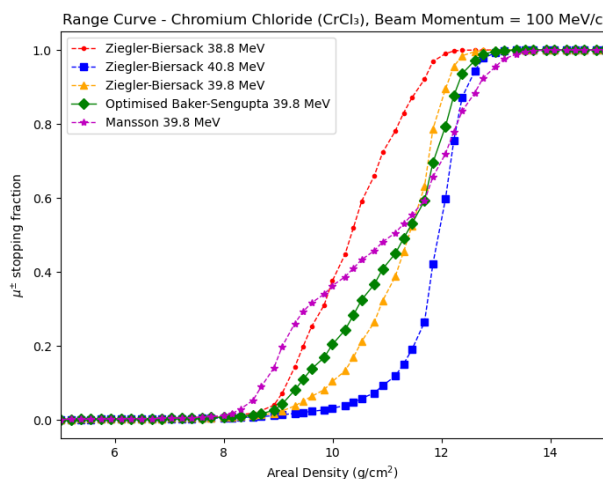


Figure 24: Range curve for different models from the PSI experiment with stopping fraction plotted against areal density.

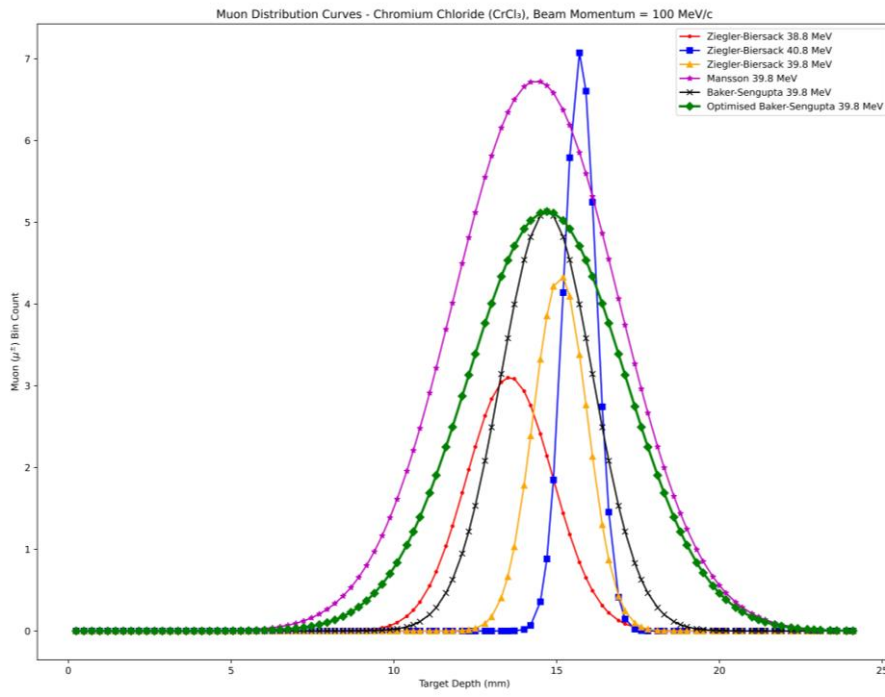


Figure 25: Muon distribution curves from different models for the PSI experiment

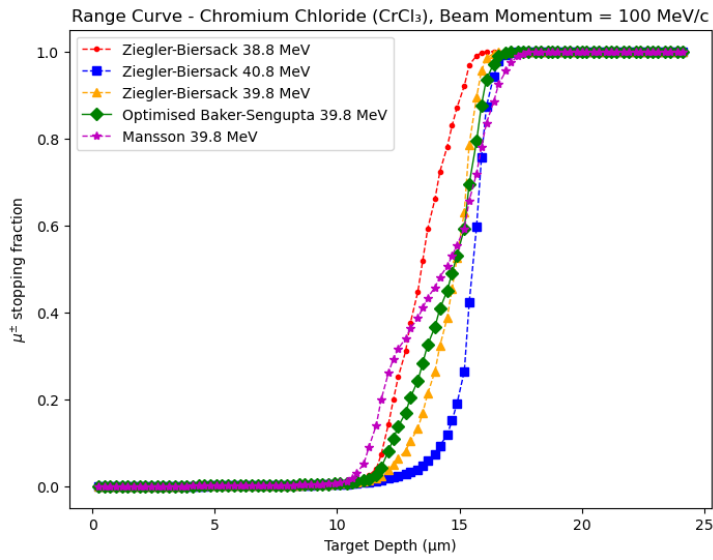


Figure 26: Range curve for different models from the PSI experiment with stopping fraction plotted against target depth.

To validate our results on the distribution and the range curves obtained from the different computational models, we analysed the beam camera images from the MuSR instrument and performed a comparative study alongside the lateral dispersion obtained from our TRIM simulations, as shown in Figure x.

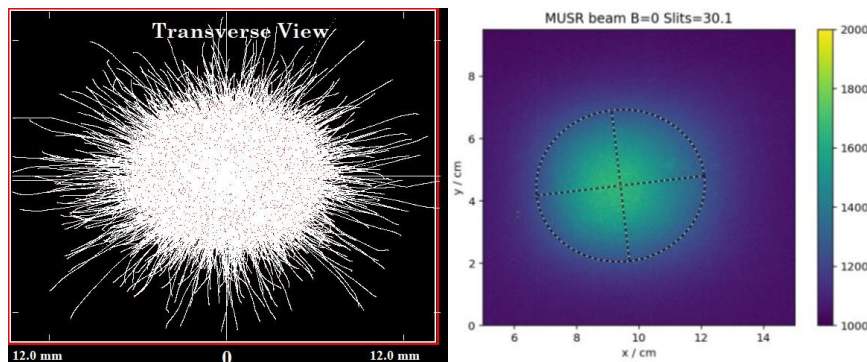


Figure 27: Beam spot / lateral dispersion plot from TRIM simulation and MuSR instrument respectively

The logic used for the image analysis involved comparing the images based on two parameters – Mean Squared Error (MSE) and Structural Similarity Measure (SSIM). From our simulations, we expect our experimental beam spot to have a uniform Gaussian distribution, as evident from the fitted three-dimensional distribution plot from TRIM in Figure x.

Ion Distribution

Ion Range = 15.3 mm Skewness = -3.465
Straggle = 1.03 mm Kurtosis = 26.837

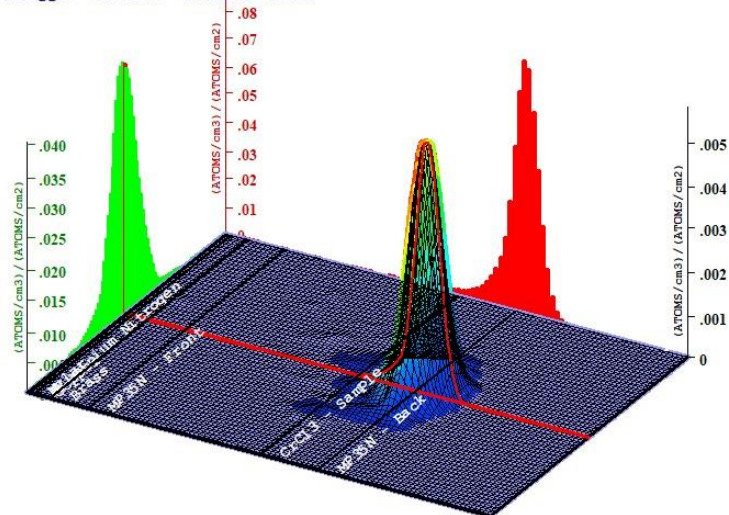


Figure 28: 3D muon distribution plot from TRIM, where the positive muons has been modelled as light hydrogen ion (H^+).

The underlying principle behind the beam spot analysis is the SSIM represents the similarity index between the experimental beam spot and the theoretical beam spot given by TRIM, while MSE can potentially provide a user with essential data on the concentration of muons in each spot on the lateral plane of the sample. The ideal result would be when the two datasets return the values of 0 and 1 for the MSE and SSIM. The expression for MSE can be given as below,

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I_{\text{experiment}}(i, j) - I_{\text{theoretical}}(i, j)]^2$$

where $I_{\text{experiment}}$ and $I_{\text{theoretical}}$ are the image tensors for the experimental and theoretical beam spots respectively. For image analysis, a value of 0 for MSE indicates similarity between two images by 100%. Any value greater than one is an indicator for less similarity between two images, as it is dependent on the average difference between pixel intensities. In the case of beam spot image analysis, the average difference between pixel intensities is equivalent to the difference between the concentration of muons that have stopped inside the sample at a given location. The MSE logic is computationally user-friendly as it is faster to compute, but it has the disadvantage of estimating only perceived errors between two images. As a result, Wang et al. proposed the use of the SSIM function to perform comparative studies between two images using computational data analysis.

$$SSIM(y, z) = \frac{(2\mu_y\mu_z + k_1)(2\sigma_{yz} + k_2)}{(\mu_y^2 + \mu_z^2 + k_1)(\sigma_y^2 + \sigma_z^2 + k_2)}$$

In the above expression for SSIM, μ_y and μ_z is the mean of the muon distribution along the y and z axes respectively, while σ_y and σ_z are the standard deviations of the respective distributions with k_1 and k_2 being random constants calculated using a MC algorithm. It is apparent that SSIM is computationally expensive leading to a slower rate of data being analysed. However, the SSIM uses local values unlike MSE (which provides you a value from a global dataset). The SSIM models the perceived change in the structural information of an image, resulting in higher accuracy in image analysis. The SSIM value varies between -1 and 1, with 1 indicating 100% similarity and vice-versa. For beam spot analysis, SSIM is helpful in comparing the beam shape and the cross-section of the experimental spot with the beam spot given by the computational simulation models.

As part of the beam spot analysis, we can notice from Figure x that the theoretical and the experimental beam spots have similar beam shape but the muon distribution and concentration at certain locations are different. This is a result of the beam having muons with multiple initial momenta / beam energy with spread as compared to the theoretical TRIM model which accounts for a beam of muons with singular initial beam energy.

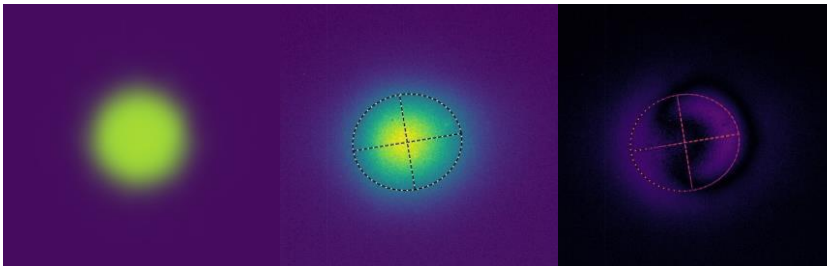


Figure 29: Image Analysis in L-R order – Beam spot from TRIM, beam spot from MuSR, difference between theoretical and experimental beam spots.

As discussed earlier, the nature of our optimisation problem is multi-objective in nature due to multiple and conflicting goals to obtain as far as the dimensions of a sample is concerned. The simplest approach used in industry is to set all goals except one as constraints. However, this approach does not address optimisation problems where two realistic, but demanding goals are of severe importance. This warrants us to use an aggregate or combined objective approach, where all the design targets are summed with weighted correction functions and minimise it. For example, if we have two goals given by functions $f_1(x)$ and $f_2(x)$, the summed function $f(x)$ is expressed as

$$f(x) = Af_1(x) + Bf_2(x)$$

where A and B are weighting constants given by the correction functions. Depending on the value of A and B , the final optimum varies. For example, $\min f_1(x) = x^2$ and $\min f_2(x) = x^2 - 4x$, the resulting function would be

$$f(x) = Ax^2 + Bx^2 - 4Bx = (A + B)x^2 - 4Bx$$

Introducing a weighting factor C given by the ratio of the weighting constants B and A , we can reduce the expression into

$$f(x) = A[(1 + C)x^2 - 4Cx]$$

The resulting function would give us a Pareto front. The best designs in the search space are referred to as “dominant” solutions. These solutions can be determined by the dominance test given by the following rules: “Solution x_1 dominates solution x_2 if x_1 is strictly better than x_2 in at least one objective and x_1 is at least equal to x_2 in all objectives.”

The resultant non-dominated solution set obtained after the dominance test is a set of solutions that are not dominated by any member of the solution set. These solutions lie within the feasible design search space / decision space and is referred to as the Pareto optimal set, and the boundary defined by the points mapped by from the solution set is called as the Pareto optimisation front.

There are different methods to obtain the combining function $f(x)$ and the Pareto front. Some of the methods to derive the combining goal function are voting schemes to rank the goals, eigenvector calculation, and fuzzy logic method. Methods to obtain the Pareto optimal front can range from response surface schemes to multi-objective population-based search, to using non-linear optimiser functions.

The three primary goals for finding the Pareto front can be stated as follows: (a) the front is well populated with feasible designs that span its full extent, (b) the designs studied by a user lies on the front i.e., it is well optimised, and (c) there is an even spread of points on the front. The nature of the computational simulation model accounts well for uncertainty quantification that might be needed to ensure robustness in an optimisation process.

In order to validate the optimisation process methodology, three separate Pareto plots were essential for a successful qualification. The first plot was a Pareto front generated by plotting asymmetry and the corresponding value for number of titanium degraders that was used during the M2-V2 experiment, where the relaxing asymmetry and number of degraders can be expressed as functions dependent on the thickness of the sample x . The second and third plots are Pareto plots comparing the predictions of the stopping fraction from different models for the M2-V2 and PSI experiments respectively. These plots have been shown in Figure 30, Figure 31 and Figure 32. It was noticed that the Optimised Baker-Sengupta outperformed all other models in all aspects – stopping fraction and depth predictions, relaxing asymmetry calculations and Ti degrader prediction – using *musrSim* results and M2-V2 data as the standard for comparison.

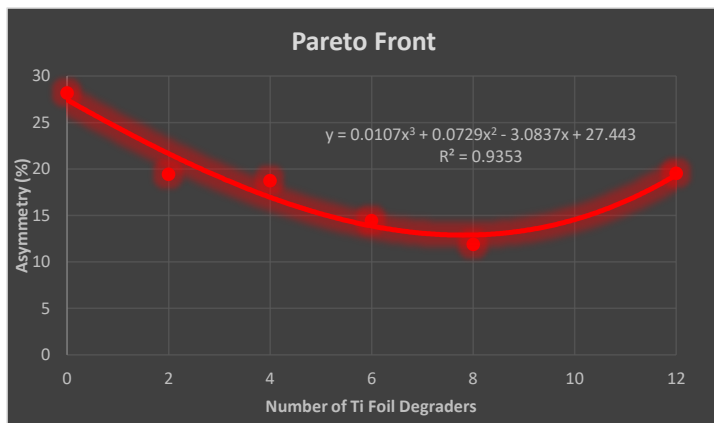


Figure 30: Muon Relaxing Asymmetry v/s Ti Degraders Pareto plot from M2-V2 experimental run.

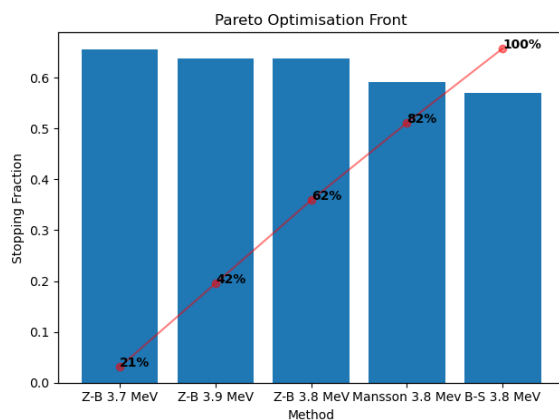


Figure 31: Pareto plot for stopping fraction prediction from different models for M2-V2 experiment.

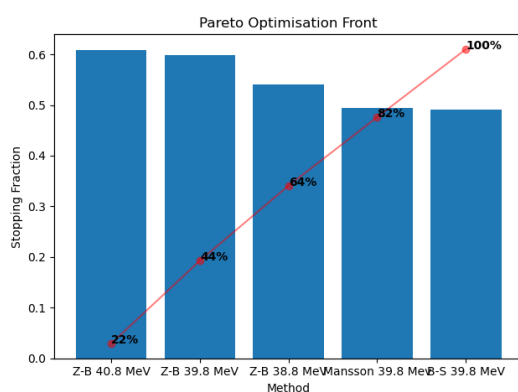


Figure 32: Pareto plot for stopping fraction prediction from different models for PSI experiment.

Conclusion

The study concludes that *musrSim* and SRIM/TRIM are software tools which complement well in a manner that provides high-fidelity simulation results for a μ SR experiment. It can also be stated with confidence that the Optimised Baker-Sengupta method (original method with straggling corrections) is the optimal computational model which could be used for nearly any muon experiments regardless of its nature of study due to its versatility, passed down by a robust software framework with *musrSim* and SRIM/TRIM being the main tools. The M2-V2 experiment effectively verified and validated all our assumptions and facts (e.g. modelling positive muons as light hydrogen ions) associated with the MESSC framework and MEDOPS optimisation process to a high degree of confidence. Since the Optimised Baker-Sengupta method is able to provide the most accurate and precise prediction for stopping fraction, target depth, range curve and other relevant data, it is the recommendation that this method be used as a standard for obtaining simulation data for μ SR experiments and also for the planning stages before a user arrives for their experimental run.

Future work involves the expansion of the optimisation process and methodology for neutron and other relevant beamlines, with applications in nuclear non-proliferation, energy, space, material science and other relevant subjects of interest. It will also involve the integration of other relevant software to the already-robust framework to simulate a muon experiment like μ SR with almost real-life accuracy with computational predictions potentially having negligible difference when compared to experimental data.

Author Contributions

The author(s)

Acknowledgements

This work was carried out at the ISIS Pulsed Neutron and Muon Source, STFC Rutherford Appleton Laboratory, UK. DS wishes to acknowledge the opportunity given by PJB and Dr Adrian Hillier that put him in contact with the μ SR technique, Dr Rhea Stewart for providing essential data and analysed beam camera images from the MuSR instrument and extends his special thanks to Dr Andy Keane and the University of Southampton in providing support with computational optimisation algorithms and sharing technical expertise on multi-objective optimisation problems.

Appendix

Appendix A: Essential Instrument and Sample Environment Information

TBD

Appendix B: Setup Variables

musrSim & *musrSimAna* – Detector Geometry, Material and Particle Definition, Physics Process, Sample and Degradation Foil, Particle Gun, Magnetic Field, Number of Particles.

SRIM & TRIM – Ion & Target Definition, Target Elements, Layer Depths and Stoichiometry, Layer Density, Damage Energies for Target Atoms.

Commented [BP(27)]: Would it be useful to have a table of useful simulation information like beam window thicknesses for instruments and cryostats so people know where to start with their simulations?

Commented [SD(28R27)]: Yes. In the sample environment section or in the software section, we can create a cross reference to one of the appendices.

Appendix C: musrSim Guide

```
# TARGET SPACE
#/musr/command construct tubs targetspace 0 5 50 0 360 G4_He 0 0 -20 log_cryostat
# SCINTILLATOR BEFORE TARGET
#/musr/command construct tubs coulombM1 0 3.0 0.1 0 360 G4_PLASTIC_SC_VINYLTOLUENE 0 0 -2 log_World norot musr/ScintSD 101
# TARGET
#/musr/command construct tubs target 0 4.0 0.215 0 360 G4_Ag 0 0 0 log_World norot dead 201
#/musr/command construct tubs targetFieldVol 0 0.5 0.015 0 360 G4_Ag 0 0 0 log_target norot dead 202
#/musr/command construct tubs vetoTarget 0 5.0 1. 0 360 G4_PLASTIC_SC_VINYLTOLUENE 0 0 1.220 log_World norot musr/ScintSD 160
#/musr/command construct tubs vetoCyl 5 6.0 3.6 0 360 G4_PLASTIC_SC_VINYLTOLUENE 0 0 -1.39 log_World norot musr/ScintSD 161
#/musr/command construct TubeWithTubeHole vetoTarget 7 9 5.1 0 360 8.2 G4_PLASTIC_SC_VINYLTOLUENE 0 0 -2.89 log_World norot musr/
ScintSD 160
#/musr/command construct TubeWithHoleAndTubeHole vetoTarget 1.5 6 5.1 0 360 4 8.2 G4_PLASTIC_SC_VINYLTOLUENE 0 0 -2.89 log_World norot
musr/ScintSD 160
#/musr/command construct tubs sampleHolder 0 4 31.39 0 360 G4_Ag 0 0 31.61 log_World norot dead 165
#/musr/command construct tubs sampleHolder2 0 7.5 108.5 0 360 G4_Ag 0 0 171.5 log_World norot dead 166
#
```

Figure x: An example of the Sample Environment section in the .mac file. The user needs to specify the geometry, the material and the location of the beam window, the sample, the sample holder and the degrader foils.

```
# Magnetic field 500G (0.05 T) in Y direction
# 1st three numbers are the half widths of the box in which the field exists
# last 6 numbers are Bx, By, Bz in Tesla, Ex, Ey, Ez in MV/m
#/musr/command globalfield TF 100 100 100 uniform 0 0 0 log_World 0 0.05 0 0 0 0
```

Figure x: An example of the Magnetic Field section in the .mac file. The user needs to define the magnetic field surrounding the target. This is an important section in the .mac file if the user is performing simulations associated with LF & TF μ SR.

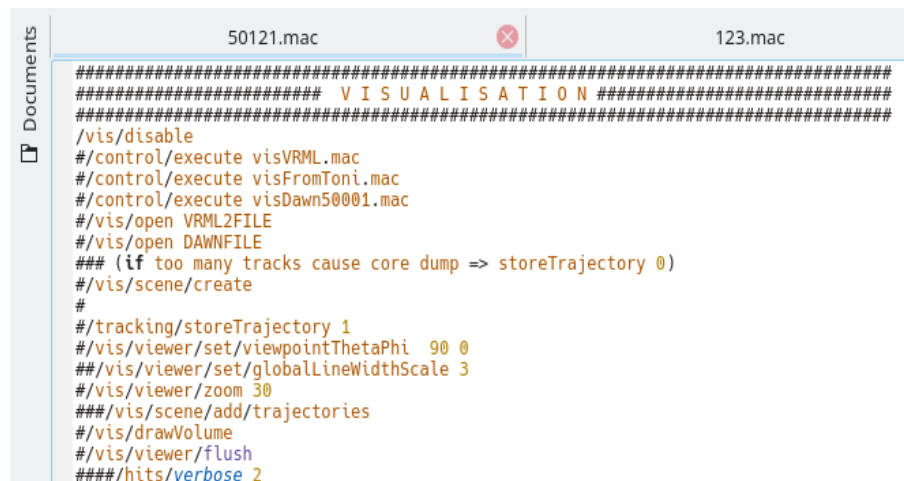


Figure x: An example of the 3D GUI visualisation section in the .mac file.

The GUI visualisation is disabled in default state but can be activated by the user by “uncommenting” one of the *control* statements. It must be noted that *musrSim* allows the user to use different GUI platforms like OpenGL to visualise the particle-target interaction within the instrument, as shown in Fig x . It is the recommendation that a maximum of 1000 muons should be simulated when visualisation is activated to prevent crashes as it can be computationally expensive.

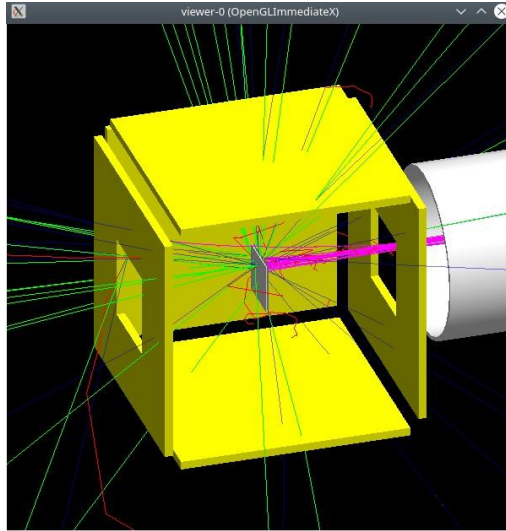


Figure x: GUI Visualisation of muon beam interacting with the sample and the sample plate, using the vis.mac file and code in *musrSim*

```
#####
##### P A R T I C L E   G U N #####
#####
/gun/vertex 0 0 -1000 mm
# FWHM 10mm ==> sigma = 10/2.354 = 4.2481mm
#/gun/vertexsigma 20 20 0 mm
#---/gun/vertexboundary: rMaxAllowed, zMinAllowed, zMaxAllowed
/gun/vertexboundary 128 -999999 999999 mm
/gun/momentum 27.0 MeV
# sigma = 3% ==> sigma 27*0.03 = 0.81
/gun/momentumsmeareing 0.81 MeV
#---/gun/momentumboundary: pMinAllowed, pMaxAllowed, dummy
/gun/momentumboundary 20 40 0 MeV
# TURTLE
#/gun/turtlefilename FOR061_2008_04_22.DAT
#/gun/turtlefilename FOR070_2008_10_17_XXII.DAT
#/gun/turtlefilename FOR077_pie3_HiField_d05_x30.dat
#/gun/turtlefilename FOR077_regglani_Jan2010_NEW.dat
#/gun/turtlefilename FOR077_regglani_Feb2010.dat
#/gun/turtleZ0position -900 mm
#---/gun/tilt: xangle, yangle, dummy
/gun/tilt 0 0.5 0 deg
#---/gun/tiltsigma: xangleSigma, yangleSigma, dummy (1 degree on 1 meter ~ 17mm)
#/gun/tiltsigma 0.2 0.2 0 deg
/gun/pitch 0.5 deg
# Spin rotated by 50 degrees upwards:
/gun/muonPolarizVector 0.766043969 0. -0.642788174
#---/gun/decaytimelimits: decayMin, decayMax, decayTime
#/gun/decaytimelimits 400 410 2197.03 ns
/gun/decaytimelimits -1 -1 2197.03 ns
#####
##### B E A M   O N #####
#####
#/run/beamOn 3000000
#/run/beamOn 10
#/run/beamOn 10000
```

Figure x: An example of the particle gun definition section in the .mac file.

In this section, the user needs to define the beam momentum in the units of MeV/c (which can be calculated using mass-energy relation for muons), the standard deviation (or “*sigma*”) to simulate muons with varying initial momentum/energy, the spin of the muon in degrees and the number of muons needed to be simulated.

Appendix D: SRIM/TRIM Guide



Figure x: SRIM user interface for calculation of stopping range of muons (modelled as hydrogen ion with 0.113 times the mass of a hydrogen ion) for a user defined target.

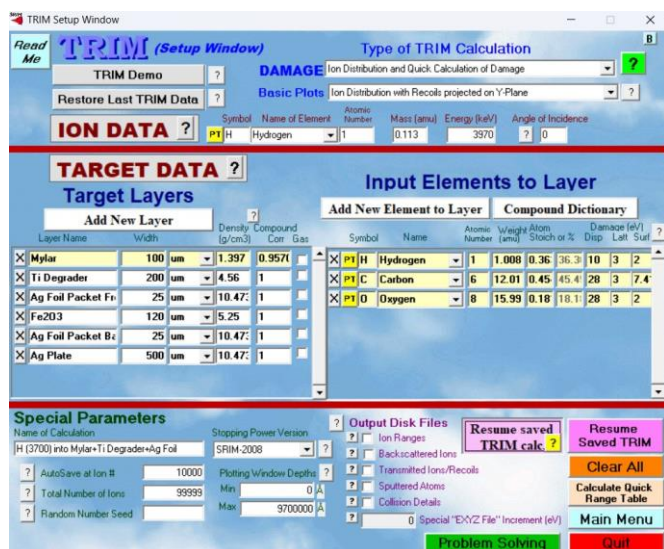
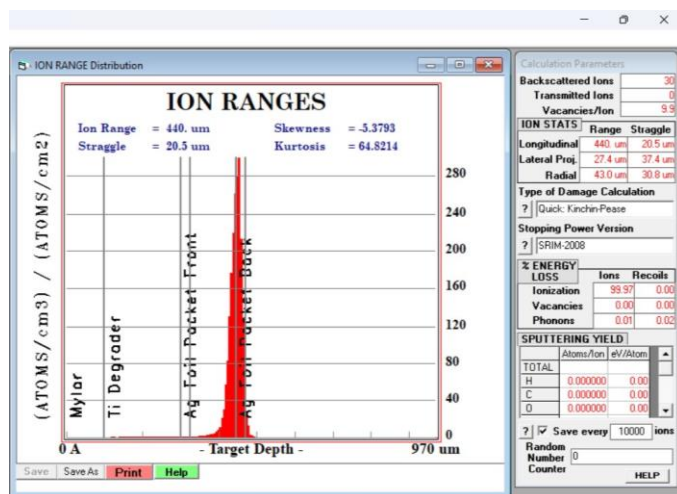
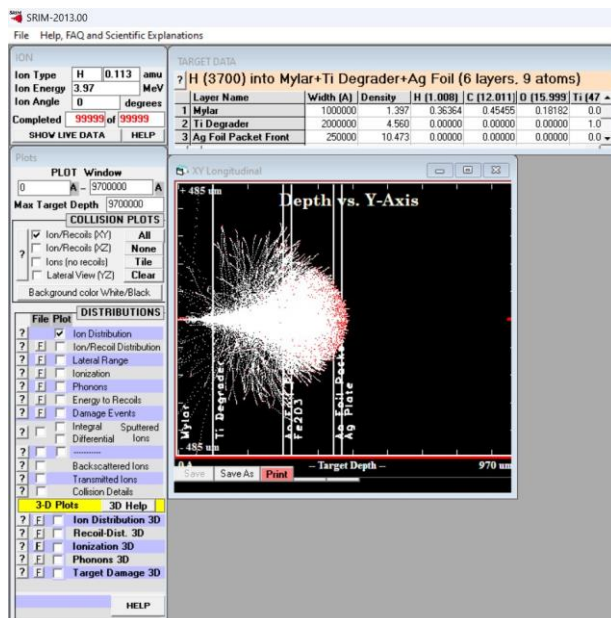


Figure x: TRIM user interface for a given beamline setup, assuming the materials are homogeneous layers of specified density and thickness. It must be noted that when a layer's material definition is a compound, the chemical composition and its associated stoichiometry needs to be defined. The interface also allows user to perform different TRIM calculations, which is subjective to the type of scientific study being conducted using μ SR experiment.



Appendix E: Important Links

TBD

References